# Boudrouss Réda | CV

Software Engineer – OCaml, TypeScript & WebAssembly – Paris

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**Languages:** French (Native), English (Professional), Arabic (Conversational)

## Profile

Software engineer completing an MSc in Computer Science (Software Science & Technology) at Sorbonne University, focused on **programming languages**, **static typing** and runtime implementation. I work in **OCaml**, **TypeScript** and Rust. I brought a large OCaml/C/C++ codebase, the MOPSA static analyzer, to the **browser** via WebAssembly, and built a **local-first**, offline-capable mobile web app backed by a low-bandwidth-optimized server.

## Projects

**Mopsa-wasm** - github.com/rboudrouss/mopsa-emcc — 2026

· Brought a large **OCaml** application to the **browser** by compiling MOPSA, a static analyzer written in OCaml/C/C++, to **WebAssembly** with Emscripten, statically linking the OCaml runtime and all native dependencies (Clang frontend, Apron, GMP, MPFR).

· Worked across the **OCaml/JavaScript boundary**, running the non-reentrant runtime in Web Workers, implementing blocking stdin over SharedArrayBuffer + Atomics for an interactive REPL, and wiring virtual-filesystem preloading to an xterm.js terminal UI.

· Shipped a fully client-side playground (zero server-side compute) and a technical write-up.

**Reactant analyzer** - github.com/rboudrouss/reactant-analyzer — 2026

· Designed and implemented a static analyzer in **Rust** (~30k lines, 500+ tests) that catches **React** hook bugs beyond the reach of syntactic linters, such as infinite render loops, derived state and unstable dependencies.

· Evaluates components in abstract domains over a dedicated CFG-based IR, flags an infinite render loop when the state fixpoint **widens**, and resolves custom hooks across files, including tsconfig path aliases in **Vite** projects.

## Experience

**Quality Analyst Engineer (apprenticeship)** - Hexaglobe (video streaming) — 2025 – 2026

· Designed complete **test strategies** for zero-coverage legacy products, reaching ~**3,000 unit / integration / E2E (Playwright) test cases** across 4 stacks (TypeScript, Python, PHP, JS).

· Built disciplined **AI-agent systems** (Claude Code) with sandboxed planner/implementer/reviewer orchestration, versioned guardrails, mandatory green tests and human review, producing ~720 integration tests in 3 days.

· Built **GitLab CI/CD** pipelines and shared CI components consumed by 4+ repositories, plus objective video-quality oracles (VMAF, freeze/black-frame detection) to validate a live encoder.

**Technical Lead, then President (2024–25)** - Association Play Sorbonne Université — 2023 – 2026

· Built a **local-first**, offline-capable mobile web app with secure authentication, a backend optimized for **low-bandwidth** clients and a database tuned for high-traffic spikes.

· Designed and operated a virtualized **infrastructure** (Docker servers, CI/CD, networking). As president, led **100+ volunteers** for a cultural event welcoming 10,000+ visitors.

## Education

**MSc in Computer Science (Software Science & Technology)** - Sorbonne University — 2024 – 2026

· Coursework: **static typing**, design & implementation of programming languages and runtimes, **concurrent & distributed programming**, static analysis by abstract interpretation.

**BSc in Computer Science** - Sorbonne University — 2020 – 2024

· Algorithms, computer architecture, operating systems, networks, databases and functional programming.